

The Internet of Things, and its implications.

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1 Abstract

The aim of this research was to answer the question:

How does the Internet of Things function, and what implications on network and societal security can be drawn from its production, operation and modern-day usage?

To do so, we establish how the "Internet Of Things" (*hence referred to as "IoT"*) is manufactured, and the flaws present in the process, as it stands today. Then, we assess the dangers it poses to the network on which such devices operate and the effects such threats pose to personal safety and daily life, while also analysing real-world case studies of such incidents.

This research was conducted using secondary sources, all of which have been duly referenced, primarily using research articles published on JSTOR.

2 The Production of IoT devices

2.1 The Diversity of IoT devices

IoT devices are extremely diverse, as their purpose is to integrate technology into our daily lives. Hence, IoT devices are manufactured by a plethora of different companies, in many different industries. Almost all industries have their feet immersed in the production of IoT, and rightly so. The Internet of Things can only exist if it exists in a variety of "things".

2.2 The effects of incompetent manufacturers

Often, manufacturers that produce such devices, are not inherently technology firms and are companies specialising in the non-IoT device they are trying to make part of the IoT, for example, a firm manufacturing an internet-connected thermostat won't necessarily be competent at managing network security. Rather, these firms often cut corners, in a bid to produce the device as

cheaply and efficiently as possible to cash in on the rising use of IoT. This is the first **fatal** flaw of IoT systems, as this practice opens gaps in the security of the IoT device and hence in its network.

2.3 Radio Communication Protocols

IoT Devices communicate using a variety of different protocols such as "Wi-Fi", "Bluetooth", "Z-Wave" or "Zigbee". While these protocols follow encryption patterns of a high standard, often adhering to the AES [*Advanced Encryption Standard*], the security risk arises due to the fact that these devices are not particularly hard to hack into. Often, ignorant manufacturers overlook cyber-security concerns such as whether the encryption key is kept secure or whether a 3rd party can connect to the device without authentication. Such was the case in the infamous "My Friend Cayla doll", which was found to be so easy to gain access to, that a hacker sitting over 20 meters away could connect to the doll's Bluetooth network, and listen to, talk to and spy on the children playing with it because the Bluetooth connection did not require any authentication to login ¹.

2.4 Vendor Specific Ecosystems vs. Open Ecosystems

However, the level of risk an IoT system poses to a household and one's life varies depending on the type of system in use. If an end-user is utilising what is known as a "vendor-specific closed loop system" ² which are devices of a specific vendor that are designed to solely communicate with other devices of that network and **not** attempt to link with devices from other vendors, then the risk is reduced, as the communication standards and methods used for such communication is often heavily protected and patented, as it is one of the vendor's trade secrets, for example, Apple's AirDrop. However, the vast majority of consumer-level IoT systems on the market are not vendor-specific and allow the integration of cheap third-party devices that meet their standards into their ecosystem. An example of such a system is Google Home, which despite having a vendor-specific array of products from thermostats to CCTV cameras, allows third-party manufacturers to integrate their hardware into the Google Home ecosystem provided they meet a list of basic requirements, allowing a user to (for instance) purchase a smart bulb from a cheap brand on Amazon, which will not have spent as much attention to detail on things like network-security and encryption, and integrate it with their Google Account and Google Home setup, enabling an adversary to gain access to the rest of one's smart-home devices, emails, contacts and beyond, all through a flaw in the cheap device.

¹C. Baraniuk, 'Call for privacy probes over Cayla doll and i-Que toys', 2016, Available at: <https://www.bbc.co.uk/news/technology-38222472> (Accessed 12 April 2024)

²S. Poudel, 'Internet of Things: Underlying Technologies, Interoperability, and Threats to Privacy and Security', *Berkeley Technology Law Journal*, vol. 31, no. 2, 2016, pp. 997-1022.

3 The Network Security Threats posed by IoT

Nowadays, the majority of home networks are accessed by consumers using "Wi-Fi" technology, allowing them to freely walk around their home while remaining connected to the central router, and hence the internet. IoT devices are hence indifferent. As they are often fitted in hard-to-reach locations like walls for CCTV, doors for doorbells etc, the use of Ethernet is ruled out in most cases, and therefore they too, send and receive **all** of their data through radio waves, by splitting the data into smaller "chunks" known as "packets". This means that during the receipt and transmission of data on a Wireless network, the data is in the open air, open to be intercepted by a hacker. Adversaries are not strangers to this, however, and the use of "packet-sniffing" attacks are becoming increasingly common.

3.1 Packet Sniffing Attacks

As discussed earlier, IoT devices present a wide array of design flaws, and weaknesses that allow an adversary to exploit the device to collect data and cause damage across a wide range of scenarios, one of which is through one's personal network. If a hacker gains access to even **one** device on the network, they can gain access to the entire network. This means they can carry out "packet-sniffing" where the hacker will essentially sit on the network, monitoring each packet as it gets sent across from device to router, until they find a packet that contains data that is useful for their intent and purpose. This can be carried out regardless of whether data is encrypted or not, as hackers can often spot patterns in the packets being transmitted.

3.1.1 Packet Dropping

While this trick may not constitute the title of a form of "attack", it certainly plays a vital role in allowing the compromising of IoT devices. Packet Dropping is where an adversary selectively places packets on the network that prevent smart-home systems from alerting the user of its use.³ This allows hackers to effectively seal the mouth of one's home IoT setup from making any noise while it is used for spying, which it would have done otherwise, in the form of notifications to the end-user device through their installed companion app.

3.2 Network "worm" attacks

However, adversaries are by no means limited to simply "sniffing" through data packets silently in the hope of a big break. Hackers can often, upon gaining ac-

³Omid Setayeshfar, Karthika Subramani, Xingzi Yuan, Raunak Dey, Dezhi Hong, In Kee Kim, Kyu Hyung Lee, 'Privacy invasion via smart-home hub in personal area networks', Pervasive and Mobile Computing, vol. 85, 2022

cess to one node on a network, release what is known as a "Worm". A "Worm" is a type of malware that self-replicates and spreads itself across the network, without the need for a trigger. This means that as soon as a worm reaches a node, it will begin replicating and spreading without the need for a user to press on a button or any other form of trigger, which its parent malware "The Virus" does. This makes them extremely volatile, as the worm itself can be any form of malware, such as ransomware that encrypts and locks your PC's files demanding a payment, to a secret key logger, that records and sends every keystroke to the adversary. Worms do not usually require a lot of processing power to begin spreading. Hence, IoT devices combined with their lack of a mainstream operating system, make a perfect proxy for the worm to enter the network without being detected⁴.

3.2.1 DDOS Attacker Farms

Hackers are not always interested in causing damage to the victim they have hacked. They are often more ambitious, targeting high-value organisations such as tech companies or banks. One of the methods by which they can achieve this goal is through a DDOS [*Distributed Denial Of Service*] attack, where an adversary floods a request server with an overwhelming amount of requests, causing it to overload and break down, and hence "deny" service to other, genuine, users. Hackers, after gaining access to multiple home networks through the security flaws of IoT, can install malware on PCs connected to it, allowing them to utilise their processing power, and connection to the internet, to flood requests at extraordinary rates, essentially creating a "PC Army", where the soldiers' owners are unaware of what their PC is used for.

4 The threats posed by IoT on individual safety and society

The effects of IoT's design failures as well as its widespread usage on network security are apparent, however, its effects on one's safety as an individual and society as a whole are not as obvious.

4.1 Spying and Surveillance

The use of IoT has placed computers in places they would have never been before. From arching over our living rooms to peering over us in our bedrooms, IoT's sheer scale within the household gives it a unique status in the eyes of

⁴S. Poudel, 'Internet of Things: Underlying Technologies, Interoperability, and Threats to Privacy and Security', Berkeley Technology Law Journal, vol. 31, no. 2, 2016, pp. 997-1022.

adversaries, who can use the devices for spying.

4.1.1 Governmental Surveillance

Governments can use IoT to carry out surveillance on their citizens, often under the guise of "crime prevention". While this does carry many benefits, including the potential to trace and catch drug criminals/gangs via the use of doorbell cameras or smartwatch GPS tracking, it also carries a negative impact on society. A society, where every internet-connected device is being used for building an image of our daily lives, and our whereabouts creates a very uncomfortable atmosphere to live in, as well as distrust between a nation and its government, about why it needs to enact such measures, and how they handle any data collected.⁵

4.1.2 Inference Intelligence

As discussed earlier, hackers are not in need of fully decrypted, plain-text data to carry out their exploits. Even the mere presence of a data packet on a network is enough for a hacker to get to work, and hence the easy gateway IoT provides to such data, acts as a huge security risk. This was demonstrated during research conducted jointly by the University of Chicago and the University of Georgia, where scientists developed a system called "*Chatterbox*" which:

*"passively eavesdrops on encrypted network traffic from the hub and leverages machine learning techniques to classify events and states of smart-home devices"*⁶

Their research showed that

the attacker can successfully disclose smart-home devices' behaviours with over 89 per cent of recall and F1-score.

This shows that a hacker does not require a decryption key to extract valuable information. By classifying packets sent across the network from particular devices into categories, using the power of AI, a hacker can pinpoint many personal details about you and your routines.

Such data can prove fatal for one's privacy and safety in their own homes, as adversaries can use this information to plan when to target your house with a burglary, and which entrance to enter through, as just **one** example, all without the use of any decryption at all, and all due to security flaw in an IoT device.

⁵J. L. Williams, 'Privacy in the Age of the Internet of Things', Human Rights, vol. 41, no. 4, 2016, pp. 14-22.

⁶Omid Setayeshfar, Karthika Subramani, Xingzi Yuan, Raunak Dey, Dezhi Hong, In Kee Kim, Kyu Hyung Lee, 'Privacy invasion via smart-home hub in personal area networks', Pervasive and Mobile Computing, vol. 85, 2022

4.1.3 Secret Recording

The discreetness of IoT devices owing to their incorporation into common household items like lights, switches, doorbells etc, allows hidden cameras to be fitted into them, such that a common victim will not notice it. Footage can be recorded in secret from any room in the house, with the simple placing of a lens into a common IoT device, designed to feed the video directly into the hacker's storage disk. This not only allows surveillance, but its use in privacy-centric locations like toilets and bedrooms means that adversaries can record potentially explicit footage which can then be shared on the dark web or used for blackmail ⁷. However, cameras do not need to be hidden as hackers can also utilise weakly secured IoT security cameras, which are trusted and are not concealed at all, allowing them to film, without the risk of being caught, completely removing any element of privacy in one's home.

4.2 Distress Attacks

4.2.1 The use of embedded microphones

Embedded Microphones can sometimes be used to instil fear into innocent victims. Baby Monitors are a special mention in this regard. There are thousands of cases of baby monitors being compromised and used to create real-life hell for parents and children alike. In one case, a family heard a man speaking to their infant in their bedroom at night. Upon arriving in the room, the hacker made a threat to kidnap their other child in the next-door room, while claiming to be inside it, creating unimaginable amounts of distress and panic for them. In another case, a family heard the swear words being screamed through their baby monitor after it was hacked. ⁸. This is as large of a safety threat as there can be for a child's safety and shows the extent to which IoT places individuals at risk of harm.

4.3 IoT Medical Devices

IoT's use in medical scenarios poses a significantly more brutal potential for damage to individual safety and well-being, due to its direct connection to the human body.

⁷C. Chirathamjaree and R. Gururajan 'Potential Security Breaches and Threats in a Wireless Computing Environment', Journal of Information Warfare, vol. 1, no. 2, 2001, pp. 68-75.

⁸J. Nakutavičiūtė, 'Hacker terrorizes family by hijacking baby monitor', 2018. Available at: '<https://nordvpn.com/blog/baby-monitor-iot-hacking/>' (Accessed 12 April 2024).

4.3.1 Cardiac Implants

The US FDA released a report⁹ describing how "St Jude Medical" cardiac implants can be easily hacked due to a major security flaw. This device was directly connected to hundreds of patient hearts, often acting as a pacemaker and therefore having direct control over their heartbeat. The hacking of such a device as a result of its security flaws, could result in the hacker being able to administer shocks to the patient's heart, by increasing the heart rate beyond the natural limit, potentially killing them.

4.3.2 CGM Devices

More devices of such nature are being introduced to the market regularly. CGM (*Continuous Glucose Monitoring*) devices are one such example, where a patient is implanted with a sensor, that continually records the glucose levels and provides readings on the person's smartphone, via the use of Bluetooth. This poses a huge risk as if a hacker gains access to the device, they can manipulate it to send false readings, such as readings showing abnormally high blood sugar, triggering the patient to inject insulin even when they don't need to or triggering the inbuilt insulin pumps on some devices to pump insulin into the blood, potentially taking someone's life.

4.4 Societal Mindset Impact

Naturally, the combination of all these factors leads to a great impact on a society's overall mentality. People live in fear that the very cameras they installed to protect them, could be used to watch them resulting in a "stranger" feeling within their own homes, as they feel they cannot do everything they want to, in the presence of all of this technology. It also creates reluctance in the use of potentially life-changing medical equipment that could turn someone's life around for the better, as they fear that the potential incompetence of the manufacturers could very well take their lives.

5 Outlook

IoT devices have, and will continue to bring, much benefit to our daily lives. Truly, it is not the underlying concept of IoT that poses a threat, but rather its implementation and usage. Changes must be implemented into the design and production of IoT to alleviate the negative implications it presents. Among many changes crucial to the IoT's sustainable future is the introduction of new

⁹S. Larson, 'FDA confirms that St. Jude's cardiac devices can be hacked', CNN Business, 2017, Available at: '<https://money.cnn.com/2017/01/09/technology/fda-st-jude-cardiac-hack/>' (Accessed 12 April 2024).

standards and protocols for the production of IoT devices that set basic requirements a company must fulfil to make IoT devices including a mandatory dedicated cybersecurity team.

Overall, IoT presents a new reality for the world and truly brings with it a great amount of benefits, IoT truly is as close as computers have come to our lives and physical selves as they have ever been, and therefore its production and use must be regulated and protected, as much as we regulate and protect our own lives.

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